TCC70xx MCU BSP

API Specification

for Serial-NOR Flash Memory Controller

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the Serial-NOR Flash Memory Controller device driver (hereinafter referred to as SFMC driver) of TCC70xx MCU BSP. For more detailed information on SFMC, refer to Chapter 7 Serial-NOR Flash Memory Controller (SFMC) in "*TCC70xx Full Specification*". [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

SFMC	Serial-NOR Flash Memory Controller
SNOR	Serial-NOR Flash Memory
SPI	Serial Peripheral Interface
GDMA	General Direct Memory Access. Refer to Chapter 14 General Direct Memory Access (DMA) Controller in *"TCC70xx Full Specification"*. [1]
DTR	Dual Transfer Rate
FWDN	Firmware Downloader

3 SOURCE FILES

3.1 Location of Source Files

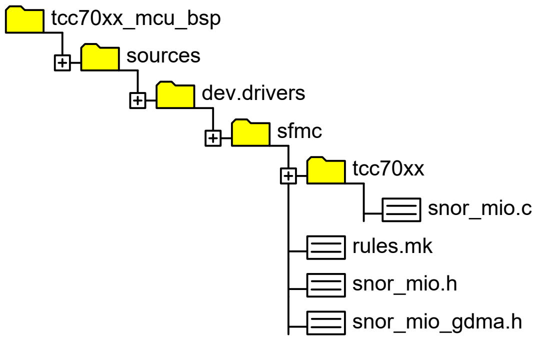

Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- snor_mio.c
  - SFMC driver source
- rules.mk
  - Rules used for build
- snor_mio.h
  - External APIs header to access SNOR
- snor_mio_gdma.h
  - Defines for using GDMA

4 CONSTANTS

4.1 SNOR_MIO_GDMA

Constant value for using GDMA

Declaration

```
#define SNOR_MIO_GDMA 1
```

Description

Value	*Description*
SNOR_MIO_GDMA	*TBD*

Remark

None

4.2 SPI Mode

These definitions indicate the properties of signals such as commands, addresses, read data, write data, and dummy.

Declaration

```
#define IO_NUM_NC        0xF
#define IO_NUM_SINGLE    0x0
#define IO_NUM_DUAL      0x1
#define IO_NUM_QUAD      0x2
#define IO_NUM_OCTA      0x3
```

Description

Value	Description
IO_NUM_NC	The corresponding signal is not output.
IO_NUM_SINGLE	The corresponding signal is output or input as Single I/O.
IO_NUM_DUAL	The corresponding signal is output or input as Dual I/O.
IO_NUM_QUAD	The corresponding signal is output or input as Quad I/O.
IO_NUM_OCTA	The corresponding signal is output or input as Octa I/O.

Remark

None

4.3 DTR Mode

These definitions enable or disable the DTR mode.

Declaration

```
#define D_DTR_DISABLE     0x0
#define D_DTR_ENABLE      0x1
```

Description

Value	Description
D_DTR_DISABLE	Disable DTR mode
D_DTR_ENABLE	Enable DTR mode

Remark

None

4.4 NOR_FLASH_BASE_ADDR

Base address for random access used when reading SNOR area

Declaration

```
#define NOR_FLASH_BASE_ADDR  (0x40000000)
```

Description

Value	***Description***
NOR_FLASH_BASE_ADDR	*Refer to Chapter 2 Address Map in “**TCC70xx Full Specification**”. [1]*

Remark

None

4.5 SFMC_BASE_ADDR

Base address of SFMC

Declaration

```
#define SFMC_BASE_ADDR    (0xA0F00000)
```

Description

Value	***Description***
SFMC_BASE_ADDR	*Refer to Chapter 2 Address Map in “**TCC70xx Full Specification**”. [1]*

Remark

None

5 STRUCTURES

5.1 snor_product_info_t

Structure for table of the SNOR device information

Declaration

```
typedef struct SNOR_MIO_PRODUCT_INFO
{
    int8 *     name;
    uint8      ManufID;
    uint16     DevID;
    uint32     TotalSector;
    uint16     cmd_read;
    uint16     cmd_read_fast;
    uint16     cmd_write;
    uint16     flags;
} snor_product_info_t;
```

Description

Value	Description
name	Name of the SNOR device
ManufID	Manufacturer's JEDEC standard ID
DevID	Device ID by manufacturer
TotalSector	Total sector number of SNOR device
cmd_read	Read command to be sent to SNOR device
cmd_read_fast	Fast read command to be sent to SNOR device
cmd_write	Write command to be sent to SNOR device
flags	Characteristics of SNOR device

Remark

None

5.2 code_table_info_t

Structure for using the CODE_TABLE register of the SFMC. For more details on CODE_TABLE, refer to Chapter 7.4 in "*TCC70xx Full Specification*". [1]

Declaration

```
typedef struct
{
    uint32 offset;
    uint32 size;
} code_table_info_t;
```

Description

Value	Description
offset	*Location of the command set in the CODE_TABLE register*
size	*Size of the command set in the CODE_TABLE register*

Remark

None

5.3 snor_mio_drv_t

Structure for SNOR device control

Declaration

```
typedef struct SNOR_MIO_DRV
{
    const int8 *      name;
    uint8             ManufID;
    uint16            DevID;
    uint8             shift;
    uint16            flags;
    uint16            current_io_mode;
    uint16            max_read_io;

    uint32            size;
    uint32            page_size;
    uint32            sector_size;
    uint32            sector_count;
    uint32            erase_size;

    uint16            cmd_read;
    uint16            cmd_read_fast;
    uint16            cmd_write;
    uint16            erase_cmd;

    uint16            dt_mode;

    code_table_info_t sfmc_buf;
    code_table_info_t sfmc_addr;
    code_table_info_t rdid;
    code_table_info_t rdsr;
    code_table_info_t wrsr;
    code_table_info_t rdcr;
    code_table_info_t wrcr;
    code_table_info_t en4b;
    code_table_info_t ex4b;
    code_table_info_t ear_mode;

    code_table_info_t write_enable;
    code_table_info_t write_disable;
    code_table_info_t read;
    code_table_info_t read_fast;
    code_table_info_t write;
    code_table_info_t blk_erase;
    code_table_info_t sec_erase;
    code_table_info_t chip_erase;

    code_table_info_t rdcr1;
    code_table_info_t rdcr2;
    code_table_info_t rdsr1;
    code_table_info_t rdsr2;
    code_table_info_t wrsr1;
    code_table_info_t wrsr2;
    code_table_info_t individual_lock;
    code_table_info_t individual_unlock;
```

```
    code_table_info_t rdab;
    code_table_info_t wrab;

    uint32            iSFMC_REG_TIMING;
    uint32            iSFMC_REG_DELAY_SO;
    uint32            iSFMC_REG_DELAY_CLK;
    uint8             uiDataBuffer[0x1000];
} snor_mio_drv_t;
```

Description

Value	Description
name	Name of the SNOR device
ManufID	Manufacturer's JEDEC standard ID
DevID	Device ID by the manufacturer
shift	Not used
flags	Characteristics of SNOR device
current_io_mode	Current SPI mode ■ IO_NUM_NC ■ IO_NUM_SINGLE ■ IO_NUM_DUAL ■ IO_NUM_QUAD ■ IO_NUM_OCTA
max_read_io	Not used
size	Total size of SNOR device
page_size	Page unit size of SNOR device to be written
sector_size	Not used
sector_count	Not used
erase_size	Erase unit size of SNOR device
cmd_read	Read command
cmd_read_fast	Fast read command
cmd_write	Write command
erase_cmd	Erase command
dt_mode	Double Transfer Rate (DTR) mode ■ D_DTR_ENABLE ■ D_DTR_DISABLE
sfmc_buf	Location of data buffer area in the CODE_TABLE register for read/write command operation

Value	Description
sfmc_addr	Location of address buffer area in the CODE_TABLE register for read/write/erase command operation
rdid	Command set area for READ ID command in the CODE_TABLE register
rdsr	Command set area for READ STATUS REGISTER command in the CODE_TABLE register
wrsr	Command set area for WRITE STATUS REGISTER command in the CODE_TABLE register
rdcr	Command set area for READ CONFIGURATION REGISTER command in the CODE_TABLE register
wrcr	Command set area for WRITE CONFIGURATION REGISTER command in the CODE_TABLE register
en4b	Command set area to enter 4-byte Address mode in the CODE_TABLE register
ex4b	Command set area to exit 4-byte Address mode in the CODE_TABLE register
ear_mode	Command set area to enter Extended Address Register mode in the CODE_TABLE register
write_enable	Command set area to enable write operation in the CODE_TABLE register
write_disable	Command set area to disable write operation in the CODE_TABLE register
read	Command set area for READ command in the CODE_TABLE register
read_fast	Command set area for READ FAST command in the CODE_TABLE register
write	Command set area for WRITE command in the CODE_TABLE register
blk_erase	Command set area for BLOCK ERASE command in the CODE_TABLE register
sec_erase	Command set area for SECTOR ERASE command in the CODE_TABLE register
chip_erase	Command set area for CHIP ERASE command in the CODE_TABLE register
rdcr1	Command set area for READ CONFIGURATION REGISTER command in the CODE_TABLE register It is used when there are multiple configuration registers of SNOR.
rdcr2	Command set area for READ CONFIGURATION REGISTER command in the CODE_TABLE register It is used when there are multiple configuration registers of SNOR.
rdsr1	Command set area for READ STATUS REGISTER command in the CODE_TABLE register It is used when there are multiple status registers of SNOR.
rdsr2	Command set area for READ STATUS REGISTER command in the CODE_TABLE register It is used when there are multiple status registers of SNOR.

Value	*Description*
wrsr1	*Command set area for WRITE STATUS REGISTER command in the CODE_TABLE register* *It is used when there are multiple status registers of SNOR.*
wrsr2	*Command set area for WRITE STATUS REGISTER command in the CODE_TABLE register* *It is used when there are multiple status registers of SNOR.*
individual_lock	*Command set area for INDIVIDUAL LOCK command in the CODE_TABLE register* *It is used when SNOR supports this command.*
individual_unlock	*Command set area for INDIVIDUAL UNLOCK command in the CODE_TABLE register* *It is used when SNOR supports this command.*
rdab	*Command set area for READ AUTOBOOT command in the CODE_TABLE register* *It is used when SNOR supports this command.*
wrab	*Command set area for WRITE AUTOBOOT command in the CODE_TABLE register* *It is used when SNOR supports this command.*
iSFMC_REG_TIMING	*Not used*
iSFMC_REG_DELAY_SO	*Not used*
iSFMC_REG_DELAY_CLK	*Not used*
uiDateBuffer	*Dynamic allocation area for data buffers*

Remark

None

6 APIs

6.1 SNOR_MIO_Init

Initializes the SFMC and SNOR device.

Declaration

```
sint32 SNOR_MIO_Init
(
    void
);
```

Parameters

None

Return

Value	*Description*
sint32	*0: Success* *-1: Failure*

Remark

None

6.2 SNOR_MIO_Erase

Erases the SNOR area from the address as much as the ***size*** parameter below.

Declaration

```
void SNOR_MIO_Erase
(
    uint32     address,
    uint32     size
);
```

Parameters

Value	Description
address	*[in] Start address of the area to be erased.*
size	*[in] Size in bytes. However, the size value must be a multiple of erase size, such as 4 Kbytes or 64 Kbytes.*

Return

None

Remark

None

6.3 SNOR_MIO_Write

Writes the pBuffer to SNOR area from the address as much as the length of the pBuffer.

Declaration

```
sint32 SNOR_MIO_Write
(
    uint32          address,
    const void *    pBuffer,
    uint32          length
);
```

Parameters

Value	Description
address	[in] Start address of the area to write pBuffer
pBuffer	[in] Buffer of data to be written to SNOR
length	[in] Length in bytes

Return

Value	Description
sint32	0: Success -1: Failure

Remark

None

6.4 SNOR_MIO_Read

Copies the length from the address of the SNOR area to the pBuffer by using read command.

Declaration

```
sint32 SNOR_MIO_Read
(
    uint32      address,
    void *      pBuffer,
    uint32      length
);
```

Parameters

Value	Description
address	[in] Start address of the area to be read
pBuffer	[out] Buffer of data to be read from SNOR
length	[in] Length in bytes

Return

Value	Description
sint32	0: Success -1: Failure

Remark

None

6.5 SNOR_MIO_Read_Fast

Copies the length from the address of the SNOR area to the pBuffer by using fast read command.

Declaration

```
sint32 SNOR_MIO_Read_Fast
(
    uint32      address,
    void *      pBuffer,
    uint32      length
);
```

Parameters

Value	Description
address	[in] Start address of the area to be read
pBuffer	[out] Buffer of data to be read from SNOR
length	[in] Length in bytes

Return

Value	Description
sint32	0: Success -1: Failure

Remark

None

6.6 SNOR_MIO_FWDN_LowFormat

Erases the entire area of SNOR by using chip erase command. Because NOR flash memory has an Erase-Before-Write structure, it does not require Low-Format like NAND flash memory, but this function is implemented for entire area erase function in Firmware Downloader (FWDN) mode.

Declaration

```
uint32 SNOR_MIO_FWDN_LowFormat
(
    void
);
```

Parameters

None

Return

Value	Description
sint32	0: Success -1: Failure

Remark

None

6.7 SNOR_MIO_FWDN_Read

This function is implemented for read function in FWDN mode. This function works the same as SNOR_MIO_Read function.

Declaration

```
sint32 SNOR_MIO_FWDN_Read
(
    uint32      address,
    uint32      length,
    void *      buff
);
```

Parameters

Value	Description
address	[in] Start address of the area to be read
length	[in] Length in bytes
buff	[out] Buffer of data to be read from SNOR

Return

Value	Description
sint32	0: Success -1: Failure

Remark

None

6.8 SNOR_MIO_FWDN_Write

This function is implemented for write function in FWDN mode. This function includes the erase operation, and if the data in the buffer to be written is the same as the data already written in the SNOR area, the operation is implemented such as skipping the write.

Declaration

```
sint32 SNOR_MIO_FWDN_Write
(
    uint32     address,
    uint32     length,
    void *     buf
);
```

Parameters

Value	Description
address	[in] Start address of the area to be written
length	[in] Length in bytes
buf	[in] Buffer of data to be written to SNOR

Return

Value	Description
sint32	0: Success -1: Failure

Remark

None

7 How to use SFMC Driver

The SFMC driver must be initialized before read, write, or erase operations. In SNOR, byte access is possible in the case of a read operation, but it should be noted that the size unit of the write operation and the size unit of the erase operation are different.

7.1 Initialize SFMC Driver

```
{
    SNOR_MIO_Init();
    ...
}
```

7.2 Read Operation

```
{
    ...
    sint32 ret;
    uint32 offset = 0x00020000;
    uint8  buf[SIZE];

    ret = SNOR_MIO_Init();
    ret = SNOR_MIO_Read(offset, buf, SIZE);
    ...
}
```

7.3 Erase/Write Operation

```
{
    ...
    uint32 offset = 0x00020000;
    uint8  buf[SIZE] = {0x12, 0x34, 0x56, ...};

    ret = SNOR_MIO_Init();
    /* It must be erased before writing */
    SNOR_MIO_Erase(offset, SIZE);
    ret = SNOR_MIO_Read(offset, buf, SIZE);
    ...
}
```

8 REFERENCES

[1] TCC70xx Full Specification

[2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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